THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

ANSWER KEY & MARKING SCHEME SET A

Chapters 1–4: **Exploring Substances: Acidic, Basic, and Neutral, Electricity: Circuits and Their Components, The World of Metals and Non-metals, Changes Around Us: Physical and Chemical** | Class 7 Science | Maximum Marks: 50

SECTION — A (Objective Type) — Answers — 12 × 1 = 12

Q.	Correct Option	Brief Reasoning
1	**(a)** neutralisation	gives salt and water
2	**(a)** Soap solution	feels soapy and turns red litmus blue
3	**(c)** tartaric acid	gives sour taste
4	**(c)** positive terminal	convention
5	**(d)** fuse	wire melts on overload
6	**(b)** closed	current flows
7	**(c)** acidic	turn blue litmus red
8	**(a)** Bromine	reddish-brown liquid
9	**(c)** tarnish	reaction with sulfur compounds
10	**(c)** chemical change	new substance forms
11	**(b)** physical, chemical	both occur
12	**(c)** ignition temperature	kindling point

SECTION — B (Very Short Answer) — Answers — 5 × 2 = 10

Q.	Expected Answer / Value Points	Marks
13	Blue litmus turns red in an acid (2).	2
14	The circuit is broken so current cannot flow (2).	2
15	A representation of a circuit using standard symbols (2).	2
16	Mercury and bromine (2).	2
17	Reaction with oxygen giving heat/light (1); wood and LPG (1).	1+1

SECTION — C (Short Answer) — Answers — 6 × 3 = 18

Q.	Expected Answer / Value Points	Marks
18	P and S acidic (1); Q basic (1); R neutral (1).	1+1+1
19	Cell weak (1); fused filament (1); loose connection or open switch (1).	1+1+1
20	Heat one end of a metal rod with wax beads (1); beads melt and fall in order (1); heat flows through metal (1).	1+1+1
21	Physical: melting butter, dissolving salt (1.5); chemical: burning candle, cooking egg (1.5).	1½+1½
22	Weakens iron objects (1.5); painting, oiling, galvanising (1.5).	1½+1½
	OR: Melting wax is physical (1.5); burning wick/vapour is chemical (1.5).	1½+1½
23	Stomach juice has acid (1); bile and pancreatic juice are basic and neutralise acid (2).	1+2
	OR: Bee sting is acidic, treated with a base like baking soda (1.5); wasp sting is basic, treated with vinegar (1.5).	1½+1½

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
24	(i) Heating and light effects (2). (ii) Heating/light (1). (iii) Heating (2).	2+1+2
	OR: (i) Lemon and salt solutions conduct (1). (ii) Sugar does not produce charged particles (1). (iii) Pure water has none (1). (iv) To make it conduct (2).	1+1+1+2
25	(i) Iron + oxygen + water (1). (ii) Test tubes experiment (2). (iii) Paint, galvanise, oil (2).	1+2+2
	OR: (i) Fizzing (1). (ii) Carbon dioxide (1). (iii) Turns milky (1). (iv) Yes (2).	1+1+1+2

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.

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ANSWER KEY & MARKING SCHEME SET B

Chapters 1–4: **Exploring Substances: Acidic, Basic, and Neutral, Electricity: Circuits and Their Components, The World of Metals and Non-metals, Changes Around Us: Physical and Chemical** | Class 7 Science | Maximum Marks: 50

SECTION — A (Objective Type) — Answers — 12 × 1 = 12

Q.	Correct Option	Brief Reasoning
1	**(b)** reddish-brown	soap solution turns it red
2	**(c)** lime (slaked lime)	a base
3	**(a)** acidic	acid test
4	**(a)** current heats the filament	heating effect
5	**(b)** Plastic	does not conduct
6	**(c)** direct current (DC)	flows one way
7	**(a)** carbon	fuel
8	**(c)** iron	haemoglobin
9	**(b)** mixture of metals (or a metal and non-metal)	e.g. steel, brass
10	**(d)** corrosion (chemical change)	new compound forms
11	**(c)** desirable	useful
12	**(a)** a physical change	state change

SECTION — B (Very Short Answer) — Answers — 5 × 2 = 10

Q.	Expected Answer / Value Points	Marks
13	The sting has formic acid (1); baking soda is a base and neutralises it (1).	1+1
14	Thin wire in a bulb (1); it heats and glows (1).	1+1
15	Producing a ringing sound on being struck (1); a bell (1).	1+1
16	Malleability: beaten into sheets, e.g. aluminium foil (1); ductility: drawn into wires, e.g. copper (1).	1+1
17	Cooking rice; milk turning into curd (2).	2

SECTION — C (Short Answer) — Answers — 6 × 3 = 18

Q.	Expected Answer / Value Points	Marks
18	Acid + base give salt and water (1); lime water added drop by drop to lemon juice with litmus (1); colour change shows the nature changes (1).	1+1+1
19	Connect cell terminal to bulb (1); bulb to switch (1); switch back to the other terminal (1).	1+1+1
20	Powder: non-metal, brittle (1.5); flattens: metal, malleable (1.5).	1½+1½
21	New substance (1); reversibility (1); energy changes (1).	1+1+1
22	Acid: sour, red litmus unchanged/blue to red (1.5); base: bitter, soapy, red litmus to blue (1.5).	1½+1½
	OR: Put drops on blue and red litmus (1); lemon turns blue to red, soap turns red to blue (1); sugar no change (1).	1+1+1
23	Terminals reversed or not touching contacts (1.5); circuit incomplete (1.5).	1½+1½
	OR: Make a circuit with a gap (1); place the material across the gap (1); bulb glows if conductor (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
24	(i) Lemon, vinegar, curd, tamarind, amla (2). (ii) Sour (1). (iii) Some can be harmful (2).	2+1+2
	OR: (i) Substance that changes colour with acids and bases (1). (ii) Paste, dip filter paper, dry and cut (2). (iii) Yellow in acid, red in base (2).	1+2+2
25	(i) Gold is less reactive (1). (ii) A layer formed by gases/air (1). (iii) Clean with polish (1). (iv) Lustrous and non-corroding (2).	1+1+1+2
	OR: (i) Table of 4 properties (2). (ii) Mercury, graphite, iodine (2). (iii) Show rules have exceptions (1).	2+2+1

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.

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ANSWER KEY & MARKING SCHEME SET C

Chapters 1–4: **Exploring Substances: Acidic, Basic, and Neutral, Electricity: Circuits and Their Components, The World of Metals and Non-metals, Changes Around Us: Physical and Chemical** | Class 7 Science | Maximum Marks: 50

SECTION — A (Objective Type) — Answers — 12 × 1 = 12

Q.	Correct Option	Brief Reasoning
1	**(c)** indicators	e.g. litmus, turmeric, China rose
2	**(c)** basic	helps remove grease
3	**(c)** Distilled water	pure water
4	**(b)** Cells	low voltage
5	**(b)** positive (+) and negative (−)	electrodes
6	**(c)** a conductor	a non-metal that conducts
7	**(c)** Nichrome (alloy)	high resistance, melting point
8	**(a)** galvanisation	e.g. buckets
9	**(d)** rust	iron oxide
10	**(a)** bubbles of gas	carbon dioxide formed
11	**(b)** Dissolving sugar in water	recoverable
12	**(a)** chemical change	new product

SECTION — B (Very Short Answer) — Answers — 5 × 2 = 10

Q.	Expected Answer / Value Points	Marks
13	Acetic, lactic and citric acid (2).	2
14	Conductors: copper, aluminium; insulators: rubber, plastic (2).	2
15	The shine of a surface (1); gold and silver (1).	1+1
16	Formation of crystals from a solution (1); physical (1).	1+1
17	Light produced by living organisms without heat (1); fireflies (1).	1+1

SECTION — C (Short Answer) — Answers — 6 × 3 = 18

Q.	Expected Answer / Value Points	Marks
18	Turmeric does not change colour with acids (1.5); it only shows bases (1.5).	1½+1½
19	Opens or closes the circuit (1.5); a light switch at home (1.5).	1½+1½
20	Sonorous (1.5); produce a clear ringing sound (1.5).	1½+1½
21	Food becomes unfit (1.5); refrigerate, pickle, seal (1.5).	1½+1½
22	Do not touch bare wires (1); keep water away (1); use only cells in experiments (1).	1+1+1
	OR: Rubber and wood are insulators (1.5); protect from shock (1.5).	1½+1½
23	It is malleable (1); does not corrode easily (1); is non-toxic (1).	1+1+1
	OR: Oxygen for respiration, carbon as fuel, iodine as antiseptic, sulfur in matches (3).	3

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
24	(i) Insulators prevent shock (1). (ii) Metals (1). (iii) Water conducts so risk increases (1). (iv) Switch off mains, dry hands, use insulators (2).	1+1+1+2
	OR: (i) Cell, bulb, switch in loop (1). (ii) Weak cell, fused filament, loose contact (2). (iii) Connect it to a fresh cell (1). (iv) Cell/battery (1).	1+2+1+1
25	(i) Five changes with classification (3). (ii) Desirable or undesirable (2).	3+2
	OR: (i) Definitions (2). (ii) Examples (2). (iii) New substance, gas, colour change, heat/light (1).	2+2+1

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.